Name : Aghera Deep

Roll no : 3187

Division : C

Subject : SS

Assignment – 4

Que1 >

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
#include <stdlib.h>
#include <ctype.h>
char mnemonic[3][3][10] =
{
{"1", "START", "AD"},
{"2", "EQU", "AD"}};
char symbol_table[10][2][10] = {""};
int s1 = 0;
int main()
{
int i = 0, j = 0;
int loc = 0;
int start = 0, equ = 0;
char *field, record[200], const1[10];
  char symb_loc[25];
```

```
   int n;
   char op[20];
   FILE *fr;
   clrscr();
   fr = fopen("Q1.txt", "r");
   printf("------------FILE DATA------------\n");
   while (!feof(fr))
   {
int fcnt = 0;
loc++;
fgets(record, 200, fr);
field = strtok(record, " ");
       while (field != NULL)
       {
    fcnt++;
    printf("%s \t", field);
           if (fcnt == 1)
           {
               if (strcmp(field, "$") != 0)
               {
                   strcpy(symbol_table[s1][0], field);
                   strcpy(op, field);
     sprintf(symb_loc, "%d", loc);
                   strcpy(symbol_table[s1][1], symb_loc);
     s1++;
               }
           }
           if (fcnt == 2)
           {
```

```
int found = 0;
int index;
for (i = 0; i < 3; i++)
{
    if (strcmp(mnemonic[i][1], field) == 0)
{
        found = 1;
        index = i;
        break;
    }
}
if (found == 1)
{
            char class1[10] = "";

    char mnemonic1[10] = "";
strcpy(class1, mnemonic[index][2]);
    strcpy(mnemonic1, mnemonic[index][1]);
    if (strcmp(class1, "AD") == 0)
    {
        if (strcmp(mnemonic1, "START") == 0)
        {
            start = 1;
        }
        if (strcmp(mnemonic1, "EQU") == 0)
        {
 equ = 1;
            loc--;
        }
```

```
            }
        }
    }
    if (fcnt == 3)
    {
        if (start == 1)
        {
strcpy(const1, field);
            loc = atoi(const1);
            loc = loc - 1;
            start = 0;
        }
        if (equ == 1)
        {
            char index_of_symbol[20];
            int find_index = 0;
            for (i = 0; i < s1; i++)
{
                if (strcmp(symbol_table[i][0], field) == 0)
                {
                    if (strcmp(symbol_table[i][1], " ") != 0)
                    {
                        find_index = 1;
                        strcpy(index_of_symbol, symbol_table[i][1]);
                        break;
                    }
                }
}
                    if (find_index == 1)
```

```
                {
             for (i = 0; i < s1; i++)
             {
                if (strcmp(symbol_table[i][0], op) == 0)
                {
                   strcpy(symbol_table[i][1], index_of_symbol);
                   break;
                }
 }
 find_index = 0;
   }
   equ = 0;
 }
   }
   field = strtok(NULL, " ");
}
   }
   fclose(fr);
   printf("\n\n------------");
   printf("\nsymbol table\n");
   printf("------------");
   for (i = 0; i < s1; i++)
   {
printf("\n");
for (j = 0; j < 2; j++)
{
   printf("%s \t", symbol_table[i][j]);
}
   }
```

```
    printf("\n------------\tName : Aghera Deep  Roll number : 3187  Assignment : 4 -Que1-");
    getch();
    return 0;
}
```

```
FPS 0 ------------FILE DATA------------
    $       START   101
            $       MOVEM   AREG    A
            LOOP    MOVER   AREG    A
            $       MOVER   CREG    B
            $       BC      ANY     NEXT
            NEXT    SUB     AREG    A
            LAST    STOP
            $       BC      LT      LOOP
            A       DS      1
            B       DS      1
            BACK    EQU     LOOP
            $       END

    ------------
    symbol table
    ------------
    LOOP    102
    NEXT    105
    LAST    106
    A       108
    B       109
    BACK    102
    ------------    Name : Aghera Deep  Roll number : 3187  Assignment : 4 -Que1-_
```

Que 2 >

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
char mnemonic[5][3][10] =
{
{"1", "START", "AD"},
{"2", "EQU", "AD"},
{"3", "ORIGIN", "AD"},
{"4", "LTORG", "AD"},
{"5", "END", "AD"}};
char symbol_table[10][2][10] = {""};
char lit_table[10][2][10] = {""};
```

```
int pool_table[10][2] = {0};
int s1 = 0, l1 = 0, p1 = 0, l_cnt = 0;
int main()
{
    int i = 0, j;
    int loc = 0;
    int start = 0, equ = 0, origin = 0, ltorg = 0, end = 0;
    char *field, record[200], const1[10];
    char symb_loc[25];
    int n;
    char op[20];
    FILE *fr;
    clrscr();
    pool_table[0][0] = 1;
    pool_table[0][1] = 0;
    fr = fopen("Q2.txt", "r");
    while (fgets(record, 200, fr))
    {
  int fcnt = 0; // field counter
  loc++;
  field = strtok(record, " ");
  while (field != NULL)
  {
      fcnt++;
      printf("%s \t", field);
      if (fcnt == 1)
            {
                if (strcmp(field, "$") != 0) // if field is not $ then label exist
                {
```

```
            strcpy(symbol_table[s1][0], field);
            strcpy(op, field);
    sprintf(symb_loc, "%d", loc);
    strcpy(symbol_table[s1][1], symb_loc);
    s1++;
} // if not '$'
   } // if fcnt=1
   if (fcnt == 2)
   {
int found = 0;
int index;
for (i = 0; i < 5; i++)

{
    if (strcmp(mnemonic[i][1], field) == 0)
    {
                found = 1;
                index = i;
                break;
            }
        }
        if (found == 1)
        {
            char class1[10] = "";
            char mnemonic1[10] = "";
            strcpy(class1, mnemonic[index][2]);
            strcpy(mnemonic1, mnemonic[index][1]);
            if (strcmp(class1, "AD") == 0)
```

```
                {
if (strcmp(mnemonic1, "START") == 0)
                    {
                        start = 1;
                    }
                    if (strcmp(mnemonic1, "EQU") == 0)
{
                        equ = 1;
                        loc--;
                    }
                    if (strcmp(mnemonic1, "ORIGIN") == 0)
                    {
                        origin = 1;
                        loc--;
                    }
                    if (strcmp(mnemonic1, "LTORG") == 0)
                    {
                        ltorg = 1;
                        loc--;
                        break;
}
                    if (strcmp(mnemonic1, "END") == 0)
                    {
                        end = 1;
                        loc--;
}
                }
            }
        } // if cnt=2
```

```
        if (fcnt == 3)
        {
            if (start == 1)
            {
                strcpy(const1, field);
                loc = atoi(const1);
                loc = loc - 1;
                start = 0;
            }
            if (equ == 1)
{
                char index_of_symbol[20];
                int find_index = 0;
                for (i = 0; i < s1; i++)
                {
 if (strcmp(symbol_table[i][0], field) == 0)
                    {
                        if (strcmp(symbol_table[i][1], " ") != 0)
                        {
                            find_index = 1;
                            strcpy(index_of_symbol, symbol_table[i][1]);
                            break;
                        }
                    }
                } // for complete
                if (find_index == 1)
                {
```

```
                for (i = 0; i < s1; i++)
                {
    if (strcmp(symbol_table[i][0], op) == 0)
                    {
                        strcpy(symbol_table[i][1], index_of_symbol);
                        break;
                    }
} // for complete
                find_index = 0;
            } // find_index =1 comlete
            equ = 0;
        } // if equ=1 complete
        if (origin == 1)
        {
            char origin_str[20];
            char *p;
            char index_of_symbol[20];


            int find_index = 0;
            strcpy(origin_str, field);
            p = strtok(origin_str, "+-");
            for (i = 0; i < s1; i++)
  {
                if (strcmp(symbol_table[i][0], p) == 0)
                {
                    if (strcmp(symbol_table[i][1], " ") != 0)
                    {
find_index = 1;
```

```
                    strcpy(index_of_symbol, symbol_table[i][1]);
                    break;
                }
            }
        } // for complete
        if (find_index == 1)
        {
            for (i = 0; i < s1; i++)
            {
                if (strcmp(symbol_table[i][0], op) == 0)
                {
                    char *ptr = strchr(field, '+');
                    p = (strtok(NULL, "+ -"));
if (ptr)
                        loc = atoi(index_of_symbol) + atoi(p);
                    else
                        loc = atoi(index_of_symbol) - atoi(p);
                    sprintf(symb_loc, "%d", loc);
break;
                }
            } // for complete
            find_index = 0;
        } // find_index =1 comlete
        origin = 0;
        loc--;
    }
    if (ltorg == 1)
    {
        l_cnt++;
```

```
            if (l_cnt > l1)
            {
                ltorg = 0;
p1++;
                pool_table[p1][0] = l_cnt;



                pool_table[p1][1] = 0;
                l_cnt--;
            }
  else
            {
                char *ptr;
                ptr = strchr(field, '=');
                if (ptr)
                {
                    for (i = 0; i < l1; i++)
                    {
                        if (strcmp(lit_table[i][0], field) == 0)
                        {
                            if (strcmp(lit_table[i][1], " ") == 0)
                            {
                                sprintf(symb_loc, "%d", loc);
                                strcpy(lit_table[i][1], symb_loc);
 pool_table[p1][1] = pool_table[p1][1] + 1;
                            }
                        }
                    }
                }
```

```
}
          }
          if (end == 1)
          {
              char *ptr;
              ptr = strchr(field, '=');
              if (ptr)
              {
                  for (i = 0; i < l1; i++)
                  {
                      if (strcmp(lit_table[i][0], field) == 0)
                      {
                          if (strcmp(lit_table[i][1], " ") == 0)
                          {
  sprintf(symb_loc, "%d", loc);
                              strcpy(lit_table[i][1], symb_loc);
                              pool_table[p1][1] = pool_table[p1][1] + 1;
                          }
                      }
      }
              }
          }



      } // if fcnt=3
      if (fcnt == 4) // will write literals to littable /////
      {
          // int complete=0;
          // int lable_exist=0;
```

```
            char *ptr;
            ptr = strchr(field, '=');
            if (ptr)
            {
                strcpy(lit_table[l1][0], field);
                strcpy(lit_table[l1][1], " ");
    l1++;
                // complete=1;
            }
    }
    field = strtok(NULL, " ");
} // while for all fields(tokens)
  } // eof while
  fclose(fr);
  printf("\n------------");
  printf("\nLiteral table\n");
  printf("------------");
  for (i = 0; i < l1; i++)
  {
printf("\n");
for (j = 0; j < 2; j++)
{
    printf("%s \t", lit_table[i][j]);
}
  }
  printf("\n------------");
  printf("\nPool table\n");
  printf("------------");
  for (i = 0; i <= p1; i++)
```

```c
  {
printf("\n");
for (j = 0; j < 2; j++)
{
    printf("%d      ", pool_table[i][j]);
}
  }
  printf("  Name : Deep Aghera  Roll number : 3187  Assignment : 4 -Qu2-");
  getch();
  return 0;
```

```
FPS 0          #        #        =5
               #        MOVER    AREG     =1
               #        LTORG
               #        #        =1

               #        MOVER    AREG     =2
               LAST     STOP
               #        BC       LT       BACK
               A        DS       1
               B        DS       1
               BACK     EQU      LOOP
               #        END
               #        #        =2
     ------------
     Literal table
     ------------
     #        110

              111
     =2

     ------------
     Pool table
     ------------
     1       0         Name : Deep Aghera  Roll number : 3187  Assignment : 4 -Que2-
```

```c
}
```